

Zirui (Ken) Deng

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SUMMARY

PhD candidate in Computer Science with expertise in machine learning, statistical modeling, and data-driven systems. Experienced in building deep learning models and scalable ETL data pipelines for time-series and structured data. Motivated in applying ML to large-scale recommendation, marketplace, and optimization problems.

Technical Tools: Python, MATLAB, R, Java, C/C++, AWS, SQL, Git, Linux, Bash, Excel, LaTeX

Soft Skills: Cross-functional Collaboration, Project Management, Multi-tasking, Innovation, Presentation

EDUCATION

Washington University in St. Louis

St. Louis, Missouri

PhD Candidate in Computer Science

August 2021 – May 2027 (anticipated)

• Research interest: theoretical and applied methods in information theory, quantitative modeling, and machine learning

Emory University

Atlanta, Georgia

B.S. in Applied Mathematics and Statistics & B.S. in Computer Science (Major GPA: 3.92/4.0)

August 2017 – May 2021

• Coursework: Numerical Analysis, Calculus, Linear Algebra, Optimization, Partial Differential Equations, Probability & Statistics, Data Structures and Algorithms, Database Systems, Systems Programming, Deep Learning, Machine Learning

RESEARCH EXPERIENCE

Privacy Certification for ML Models via PMC

St. Louis, Missouri

Washington University in St. Louis

August 2025 – Present

• Designing a unified privacy auditing framework leveraging Pointwise Maximum Cost (PMC), enabling quantification of membership inference risk in **LLM**, **Transformer**, and **CNNs**, achieving 18% **AUPRC** improvement.
• Developing scalable estimation pipelines (**Python**, **PyTorch**, **Pandas**) linking empirical attack metrics (**TPR/FPR**) to theoretical guarantees, supporting production-grade ML model monitoring and reducing evaluation latency by 30%.
• Analyzing adversarial strategies across models (**SVM**, **generative models**), deriving optimal inference policies and improving robustness of ML systems under distribution shift, reducing worst-case attack success by 20% (**AUROC**).

Efficient Vector Symbolic Architectures via Histogram Recovery

St. Louis, Missouri

Washington University in St. Louis

February 2025 – August 2025

• Built a neurosymbolic representation framework (**VSA**) for compositional learning, enabling efficient encoding and structured data retrieval across **Transformer** and **RNNs (LSTM, GRU)**, achieving 30% improvement in retrieval accuracy.
• Deployed polynomial-time decoding algorithms (histogram recovery, lattice decoding) to replace prior NP-hard methods, boosting 10x speedup and enabling scalable inference in high-dimensional ML representations.
• Implemented end-to-end evaluation pipelines (**Python**, **PyTorch**, **CI/CD**) to benchmark noise robustness and retrieval accuracy, supporting memory-efficient representation learning in large-scale deployments.
(Preprint link: <https://arxiv.org/abs/2511.01838>)

Private Inference in Quantized Models

St. Louis, Missouri

Washington University in St. Louis

September 2022 – May 2025

• Constructed privacy-preserving ML inference protocols for **quantized neural networks (Regressions, CNNs, etc.)** using **Pandas**, **PyTorch**, and **scikit-learn**, enabling secure inner-product evaluation with <1% accuracy degradation.
• Developed & assessed information-theoretic privacy-utility tradeoff frameworks, reducing communication complexity by ~10x while maintaining strong guarantees against inference attacks in large-scale ML systems.
• Automated scalable evaluation pipelines (**Python**, **PyTorch**) to benchmark performance across high-dimensional datasets, improving attack resistance vs. cryptographic baselines and validating robustness through stress testing.
(Publication link: <https://ieeexplore.ieee.org/abstract/document/10896693>)

Perfect Subset Privacy in Polynomial Computation

St. Louis, Missouri

Washington University in St. Louis

October 2023 – April 2025

• Designed distributed ML computation frameworks using **Reed-Muller / coded computing**, enabling privacy-preserving training and inference for **GNNs (LightGCN)** and large-scale data pipelines.
• Optimized & analyzed NLP encoding schemes with zero information leakage guarantees, achieving communication cost of lower orders of magnitude and improving scalability over naive distributed ML baselines.
• Established simulation pipelines (**Python**, **Pandas**) to evaluate robustness to system failures (e.g., stragglers), improving fault tolerance by ~25% for distributed ML workloads and large-scale data processing systems.
(Publication link: <https://ieeexplore.ieee.org/document/10619107>)

Predicting Surgical Case Duration Using Neural ODEs

Washington University in St. Louis

St. Louis, Missouri

August 2021 – November 2021

- Designed & implemented a time-series prediction pipeline using **Python**, **PyTorch**, integrating **LSTM** and **Neural ODE** models to capture both discrete and continuous dynamics in heterogeneous clinical data.
- Developed robust data preprocessing workflows for multimodal datasets (static + temporal features), improving training stability and enabling scalable experimentation across **RNN/LSTM-based** architectures.
- Optimized numerical solvers and training procedures for stability and efficiency, reducing surgical duration prediction error by 15% (**CRPS**) and improving reliability in real-world healthcare forecasting tasks.

Undergraduate Honors thesis: Fair algorithms in machine learning

Emory University

Atlanta, Georgia

May 2020 – May 2021

- Analyzed algorithmic fairness in **ML classification and clustering** models, defining fairness via algorithmic bias in ML-based decisions on sensitive variables (e.g., gender, ethnicity, sexual orientation).
- Developed and evaluated optimization-based fairness constraints (e.g., parity regularization, constrained ERM), reducing statistical parity gap by up to 40% with <5% loss in predictive accuracy.
- Evaluated fairness-utility tradeoffs under distributional shifts, showing improved out-of-sample generalization with reduced disparity by ~30% and stable predictive performance ($\leq 3\%$ drop in accuracy) on held-out test sets.

Multi-variate Neural Ordinary Differential Equations for Time-series Modeling

Remote from Emory University

Atlanta, Georgia

May 2020 – July 2020

- Designed and implemented deep learning models (**CNN**, **LSTM**, **Neural ODE**) for time-series medical data, capturing discrete and continuous dynamics and reducing forecasting error by ~12% (**MSE**) relative to baseline **RNN** models.
- Developed data augmentation and contrastive learning strategies to improve representation quality, achieving 8-10% improvement in downstream prediction accuracy and matching state-of-the-art sequence modeling benchmarks.
- Optimized training pipelines for stability and scalability (batching, learning rate scheduling, numerical stability), reducing training time by ~40% and enabling efficient experimentation across large-scale time-series datasets.

INTERNSHIPS

Data Analyst Internship

China Telecom

Nanjing, China

May 2019 – June 2019

- Developed automated data pipelines using **Python**, **Selenium**, **Pandas**, enabling large-scale web scraping (10K+ records/day) and improving data collection efficiency by ~60% for financial analysis workflows.
- Implemented validation and cleaning scripts (**Python**, **NumPy**) to ensure data integrity, reducing data inconsistencies by ~35% and lowering manual processing time by ~50%.
- Contributed to a financial robotics project by building modular data processing components, supporting near real-time ingestion and improving downstream analytics latency by ~30%.

Investment Banking Analysis Internship

Ten Billion Research Laboratory in Shanghai

Shanghai, China

May 2018 – June 2018

- Conducted quantitative market research on blockchain technologies using **Python**, **Excel**, analyzing 50+ projects/datasets and identifying key industry trends through SWOT and time-series analysis.
- Developed forecasting insights from structured and unstructured data, improving prediction accuracy of short-term market trends by ~20% compared to baseline heuristic models.
- Presented data-driven findings in cross-functional meetings, delivering actionable investment insights that informed strategy discussions and improved decision turnaround time by ~25%.

ADDITIONAL INFORMATION

Languages: Fluent in English and Mandarin Chinese

Interests: Poker; Weiqi (amateur 5-dan, national second-level athlete)